Chicago Crime Patterns and Socioeconomic Correlates

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Lecture Section: Section 4

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Research Question

This project aims to inform the targeting of public safety and social investment resources with asking two questions: (1) What are the temporal, spatial, and categorical patterns of crime across Chicago in 2024, and (2) to what extent are neighborhood crime rates associated with socioeconomic conditions (education attainment, unemployment, and median income)?

Data sources include the City of Chicago's crime incident dataset (filtered to 2024) and the 2024 American Community Survey 5-Year Estimates (ACS) for socioeconomic variables.

Approach and Coding

The analysis was organized in four stages. First, **data cleaning**: the 31 original crime types were collapsed into four categories (Violent, Property, Regulatory, Other), and ACS tables were processed to extract 11-digit GEOID codes and compute derived rates.

Second, **spatial joining**: tracts were clipped to the Chicago city boundary, and a GeoPandas `sjoin` with predicate `within` assigned each crime point to exactly one tract. Crime rate was computed as count divided by ACS population, scaled by per 100 residents.

Third, **static visualization**: Produced two temporal heatmaps, two choropleth maps of crime count and rate and three socioeconomic choropleth maps.

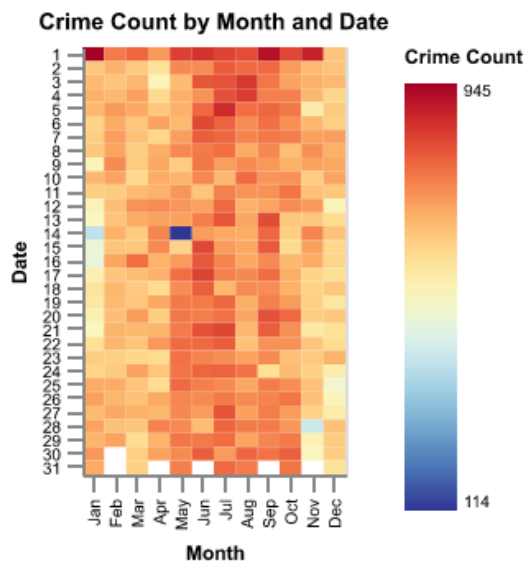
Fourth, **Streamlit app**: Users can interactively explore crime patterns and their relationship to socioeconomic variables.

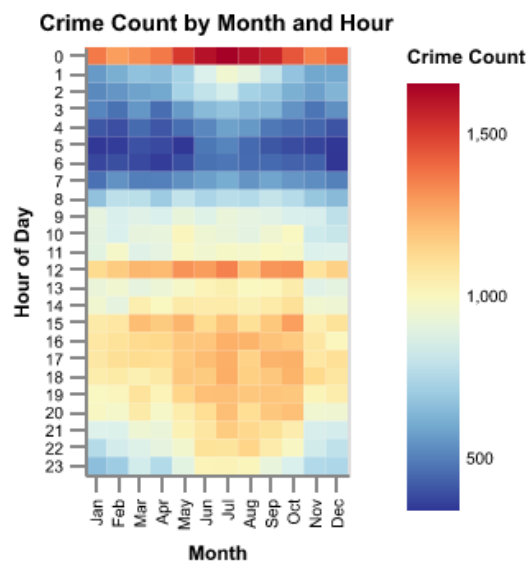
Weaknesses and difficulties encountered:

- **Temporal scope**: The crime dataset was filtered to 2024 only due to file size constraints, precluding time-series or trend analysis across years.
- **Spatial join precision**: An initial spatial join using `intersects` without first clipping tracts to the Chicago city boundary inadvertently included partial Cook County tracts and water bodies such as Lake Michigan. This was resolved by first clipping all Illinois tracts to the Chicago city boundary shapefile, then using a `within` predicate to assign each crime point to exactly one tract.
- **Correlation, not causation**: The socioeconomic analysis identifies associations between tract-level conditions and crime rates but cannot establish causal relationships.

Plots 1 & 2 — Crime Count by Month/Date and Month/Hour

The heatmaps reveal clear temporal structures. Plot 1 shows higher crime in summer months and on the 1st of each month. Plot 2 shows an evening peak (6–9 PM), especially in summer. Early-morning hours (2–5 AM) are uniformly low.



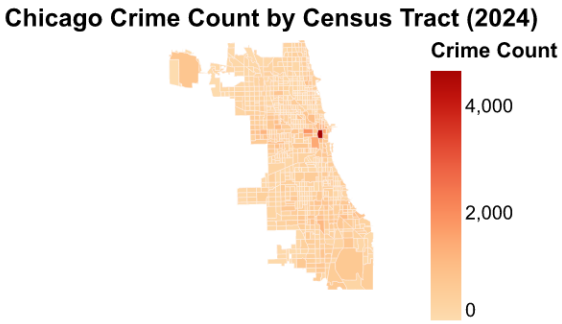


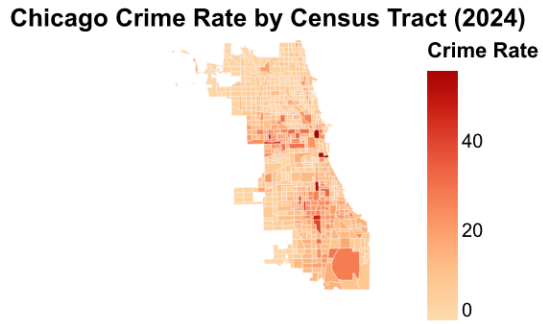
Following feedback from the presentation, we investigated several apparent anomalies:

- **Elevated counts on the 1st of each month and at 12:00 AM / 12:00 PM**: Minute- and second-level analysis confirms these are CPD default placeholder values, with exact 00:00:00 timestamps account for 9.70% of records on the 1st versus 3.31% on other dates.
- **Low counts on January 14 and November 28**: Record counts are consistent between raw and cleaned data, confirming the reductions reflect genuine behavioral changes on federal holidays (MLK Day and Thanksgiving).
- **Low count on May 14**: Of 728 raw records, only 114 remain after cleaning, as 84% are missing geographic coordinates compared to a citywide average of 99.4%. May 15 is the second-worst date by the same metric, suggesting a CPD system outage rather than a true crime drop.

Plots 3 & 4 — Crime Count and Crime Rate Choropleth Maps

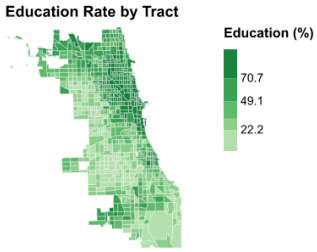
Plot 3 maps raw crime counts, where the Loop and Near North Side dominate due to high commercial and transit activity, making spatial variation elsewhere difficult to discern. Plot 4 adjusts for population (crime rate per 100 residents), producing a more evenly distributed and informative picture that highlights elevated risk in South and West Side tracts that would otherwise appear moderate in absolute terms.

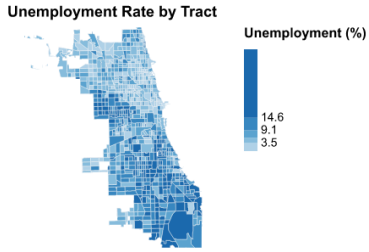


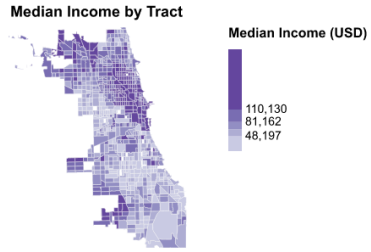


Plots 5, 6 & 7 — Socioeconomic Choropleth Maps

These three maps display the spatial distribution of education rate, unemployment rate, and median income across Chicago census tracts. All three exhibit geographic patterns strikingly similar to the crime rate map (Plot 4), with socioeconomic disadvantage concentrated in the same South and West Side tracts that bear the highest crime burden,which motivates the correlation analysis in the Streamlit app.







Streamlit App

The interactive dashboard is organized into two pages with a sidebar.

Page 1: Crime Distribution displays a choropleth map of crime rates per 100 residents and a bar chart of crime counts by category, filterable by year, month, hour, and crime type. The South and West Side tracts bear the highest crime rates, while the Loop dominates in absolute counts due to commercial activity.

Page 2: Socioeconomic Correlates presents scatter plots with OLS trendlines showing crime count and crime rate against a user-selected factor: education rate, unemployment rate, or median income. Tracts with greater socioeconomic disadvantage consistently show higher crime rates, corroborating the spatial patterns in Page 1.

These findings suggest that public safety interventions and social investment programs would have the greatest impact if targeted at South and West Side tracts, particularly during summer months and evening hours, where both crime rates and socioeconomic disadvantage are most concentrated.